

Credit Card Spending Analysis December 2019 vs. January 2020

David Riego · Statistical Computing · UA&P · May 2026

1 Abstract

Holiday shopping is widely assumed to push December credit-card spending well above January's post-holiday lull. This study tests that assumption on **192,327 simulated transactions** using both a Welch two-sample t -test and a 10,000-iteration bootstrap. Both methods converge: the **\$1.11** December–January mean difference is statistically indistinguishable from zero ($p = .121$ and $.128$). The expected "December effect" on transaction amount does not appear in these data.

2 Background

- "December = peak spending" is a default assumption for retailers and card issuers.
- The intuition is rarely tested at the transaction level.
- This study tests it with two independent inferential approaches.

Research question

Is the mean credit-card transaction amount in December 2019 significantly different from January 2020?

3 Hypotheses

NULL · H_0	ALTERNATIVE · H_A
$\mu_D = \mu_J$	$\mu_D \neq \mu_J$
Means are equal.	Means differ.

Two-sided test, $\alpha = .05$.

4 Data & Cleaning

Source. Kaggle (Choksi, 2020); simulated credit-card transactions. Raw file: 1,048,575 rows $\times$ 24 columns.

140,468 December 2019 (73.0%)	51,859 January 2020 (27.0%)
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Cleaning steps

- Convert Excel serial dates $\rightarrow$ datetime.
- Filter to December 2019 + January 2020 only.
- Remove fraud (`is_fraud = 1`): 935 rows (~0.48%).
- Validate amounts: no missing or non-positive values.

Final clean sample: **192,327 legitimate transactions**.

5 Methodology

Traditional · Welch's t -test

- Two-sample, two-sided, on raw USD amounts.
- Welch correction handles unequal n .
- CLT justifies parametric inference at this sample size.

Computational · Bootstrap

- Resample each month with replacement, preserving n .
- B = 10,000** iterations.
- Percentile 95% CI; null-shifted resamples for p .
- No distributional assumption — robust to skew.

6 EDA · Boxplots

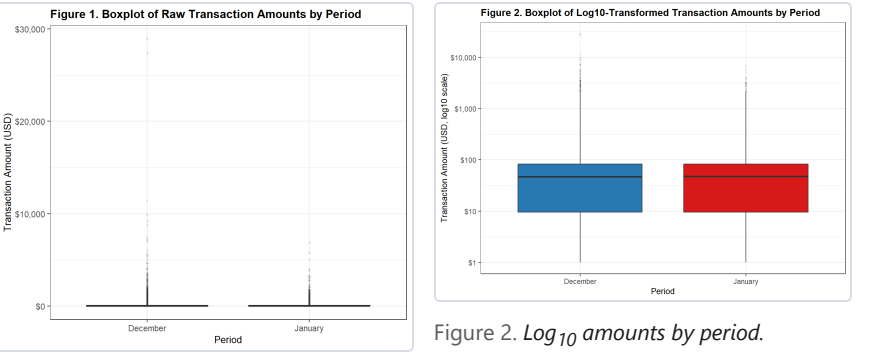


Figure 1. Raw amounts by period.

Raw scale is crushed near \$0 by extreme outliers. The log view shows **nearly identical medians, IQRs, and spread** between the two months.

7 EDA · Histograms

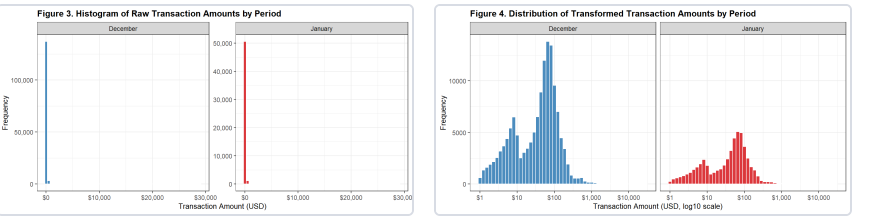


Figure 3. Raw amounts.

Figure 4. Log₁₀ amounts.

Severe right skew on raw scale. The log view reveals a **bimodal pattern at ~\$10 and ~\$100** — present identically in both months.

8 Descriptive Statistics

Statistic	December 2019	January 2020
n	140,468	51,859
Mean (USD)	\$68.22	\$67.11
Median (USD)	\$47.16	\$47.38
SD (USD)	\$175.15	\$122.82
Min / Max	\$1.00 / \$28,948.90	\$1.00 / \$6,880.74
Q1 / Q3	\$9.53 / \$82.43	\$9.57 / \$82.80
IQR	\$72.90	\$73.23
Skewness	70.98	15.92

Means differ by only **\$1.11**; medians by less than \$1 — strikingly similar centers despite severe right skew.

9 Assumption Checks

Shapiro-Wilk (normality)

Group	W	p
December	0.41	< .001
January	0.57	< .001

Levene (equal variance)

$F(1, 192325) = 2.20, p = .138 \rightarrow$ variances equal.

Normality fails — but CLT rescues the t -test at $n > 50,000$. Bootstrap provides an independent, assumption-free check.

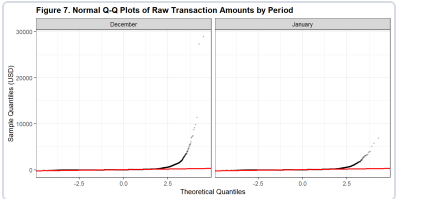


Figure 5. Normal Q-Q plots by period.

10 Result · Welch's t -test

t = 1.55 df = 131,580	p = .121 two-sided
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[−\$0.29, \$2.51]
95% CI for $\mu_D - \mu_J$

- Observed mean difference: **\$1.11**.
- $p > .05$ and CI includes zero $\rightarrow$ **fail to reject H_0** .

11 Result · Bootstrap

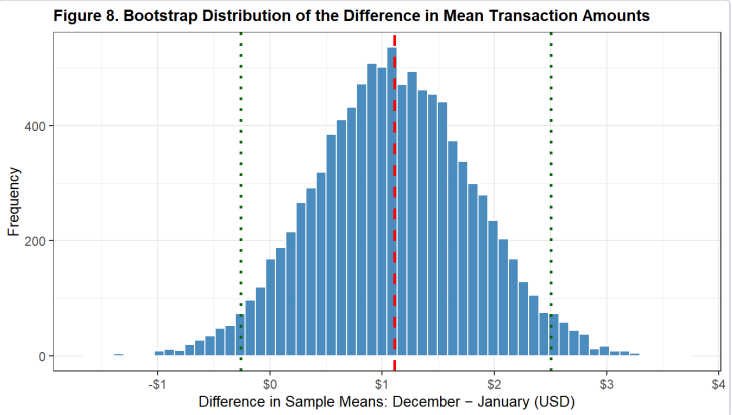


Figure 6. Bootstrap distribution of mean difference (Dec – Jan), $B = 10,000$.

p = .128 $B = 10,000$	[−\$0.26, \$2.50] 95% percentile CI
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- Distribution centered just above \$1; CI brackets zero.
- Fail to reject H_0** — same verdict as the t -test.

12 Synthesis · Triangulation

Quantity	t-test	Bootstrap
Mean diff	\$1.11	\$1.108
p-value	.121	.128
95% CI	[−\$0.29, \$2.51]	[−\$0.26, \$2.50]
Decision	Fail to reject	Fail to reject

p -values differ by **.007**; CIs nearly identical. Both methods converge — exactly what the CLT predicts at this n . Agreement strengthens the conclusion.

13 Belief · Method I Trust

Primary: Bootstrap. Confirmatory: t -test.

- Data is severely right-skewed (skew = 70.98 in December) — exactly where parametric tests are riskiest.
- Bootstrap is distribution-free and does not rely on the CLT.
- The t -test is valid *here* only because n is enormous; bootstrap is the safer default.
- Because both agree, the t -test independently validates the bootstrap.

No "December effect" on transaction amount. If a holiday signal exists, it lives in transaction frequency, not amount per swipe — managers should target **volume**, not basket size. Replicate on real data before acting; this dataset is simulated.

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Statistical Computing (DS1322), University of Asia and the Pacific.
Author: **David Riego**
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Analysis in R (Welch's t -test, bootstrap $B = 10,000$); data preparation in Python (pandas).